Convolutions - The Big Picture: Unifying Probability, Signal Processing, Statistical Mechanics, and Information Theory

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Abstract

Convolutions arise naturally and independently across probability and statistics, signal processing, statistical mechanics, and information theory. In this paper we argue that these are not parallel rediscoveries of the same formula but rather a single underlying mathematical structure whose dictionary can be used as a *transport mechanism*: a non-trivial theorem proved in one field often translates, via the correspondence, into a non-trivial and sometimes previously unrecognised result in another. Central to our discussion is the Central Limit Theorem, which we prove two ways and interpret simultaneously as a fixed-point theorem (analysis), a maximum-entropy attractor (information theory), and a thermodynamic equilibrium argument (statistical mechanics). We then carry six specific "cross-field transfers" to their conclusions, showing that the Entropy Power Inequality implies a superadditive noise-floor bound for multi-stage amplifier chains, that the Fisher Information Inequality implies bandwidth irreversibility, that Berry–Esseen gives a filter-design principle, that the De Bruijn identity equates Gaussian filtering with the heat equation and makes Fisher information measurable by blurring, that Cramér's large-deviations theorem gives the exponential tail-decay rate of any cascaded filter, and that Stein's lemma in probability is the same equation as Price's theorem in signal processing.

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1 Introduction

One of the most striking features of modern mathematics is how the same operation—the *convolution*—appears, wearing different clothes, in completely different scientific disciplines. For suitable functions f and g,

$$(f * g)(x) \;=\; \int_{-\infty}^{\infty} f(t)\, g(x-t)\, \mathrm{d}t.$$

In **probability theory** this is the PDF of the sum of two independent random variables. In **signal processing** it is the output of a linear time-invariant (LTI) system to an input signal. In **statistical mechanics** it enters when combining the density of states of independent subsystems. In **information theory** it describes how noise accumulates on an additive channel.

A naive reaction is: "same integral, different notation." This paper argues for something stronger. *The correspondence is a theorem-transporting machine.* A non-trivial result—one that required real effort to prove in its home field—often maps, under the dictionary, to a result in a different field that is either new or whose proof is now trivially inherited. Section 7 is devoted entirely to making this precise through six examples.

Throughout, the Gaussian distribution plays the role of a canonical object: it is the unique fixed point of normalised convolution (Section 4), the maximum-entropy distribution under a variance constraint (Section 4), the equilibrium velocity distribution in a gas, the amplitude distribution of bandlimited white noise, and the unique distribution whose characteristic function is also Gaussian.

2 A Correspondence Dictionary

Table 1 aligns the core objects and Table 2 aligns the core tools. Every entry will be used somewhere in the paper.

Probability / Stats	**Signal Processing**	**Stat. Mech.**	**Info. Theory**
Random variable X	Signal $x(t)$	Microstate	Source symbol
PDF $f_X(x)$	Impulse response $h(t)$	Density of states	Symbol probability
Sum $X+Y$ (indep.)	Convolution $x * y$	Combined partition fn	Noise addition
Characteristic fn $\varphi_X(\omega)$	Fourier transform $\hat{h}(\omega)$	Canonical partition fn	MGF on imaginary axis
MGF $M_X(s)$	Laplace transform $H(s)$	Grand partition fn	Cumulant gen. fn
Moments $\mathbb{E}[X^k]$	Taylor coefficients of $\hat{h}$	Thermodynamic averages	Cumulants
Fisher information $J(X)$	Inverse bandwidth^2	Inverse heat capacity	Fisher information
CLT	Gaussian cascade limit	Equipartition theorem	Max-entropy attractor
Normal distribution	Gaussian pulse	Maxwell–Boltzmann	Max-entropy distribution

Table 1: Correspondence dictionary. Each row is the same mathematical object viewed from four different vantage points.

The single structural fact underlying the entire dictionary is that **exponential functions are the eigenfunctions of convolution**: $(f * e^{i\omega\cdot})(x) = \hat{f}(\omega)\, e^{i\omega x}$. Everything else follows.

Transform	**Kernel $K(x,\xi)$**	**ξ lives in**
Fourier / Characteristic fn	$e^{\pm i\omega x}$	Real ω
Laplace (engineering, one-sided)	e^{-sx}, $x \geq 0$	Complex s
MGF (two-sided)	e^{sx}, $x \in \mathbb{R}$	Real s near 0

Table 2: All transforms are the same integral against an exponential kernel. The convolution theorem—$(f * g)^\wedge = \hat{f} \cdot \hat{g}$—holds in every row.

3 The Central Limit Theorem

Theorem 3.1 (Central Limit Theorem)**.** *Let $X_1, X_2, \ldots$ be i.i.d. with mean μ and variance $\sigma^2 < \infty$. Let $Z_n = (S_n - n\mu)/(\sqrt{n}\,\sigma)$. Then $Z_n \xrightarrow{d} \mathcal{N}(0,1)$.*

3.1 Proof via Moment Generating Functions

Proof. Set $Y_i = (X_i - \mu)/\sigma$ and let $M(s) = \mathbb{E}[e^{sY}]$. Since $Z_n = n^{-1/2}\sum Y_i$ and the Y_i are i.i.d.,

$$M_{Z_n}(s) = \big[M(s/\sqrt{n})\big]^n.$$

Taylor-expanding M near 0 (using $M(0) = 1$, $M'(0) = 0$, $M''(0) = 1$):

$$M\big(s/\sqrt{n}\big) = 1 + \tfrac{s^2}{2n} + O(n^{-3/2}),$$

so $M_{Z_n}(s) = \big(1 + s^2/(2n) + O(n^{-3/2})\big)^n \to e^{s^2/2}$. The limit is the MGF of $\mathcal{N}(0,1)$; MGF convergence on a neighbourhood of 0 implies convergence in distribution (Curtiss's theorem [3]). □

3.2 Proof via Characteristic Functions

Proof. With the same Y_i, the characteristic function satisfies $\varphi_{Z_n}(\omega) = [\varphi(\omega/\sqrt{n})]^n$. Taylor expanding: $\varphi(\omega/\sqrt{n}) = 1 - \omega^2/(2n) + O(n^{-3/2})$, so

$$\varphi_{Z_n}(\omega) \;\to\; e^{-\omega^2/2}.$$

This is the characteristic function of $\mathcal{N}(0,1)$. Lévy's continuity theorem converts pointwise characteristic-function convergence into convergence in distribution. □

Remark 3.2 (Why the Gaussian is the *only* possible limit)**.** *By Marcinkiewicz's theorem [4], if the characteristic function of a distribution is* $\exp(P(\omega))$ *for a polynomial P, then P must have degree ≤ 2 and the distribution is Gaussian. Stability under convolution + rescaling forces $P(\omega) = -c\omega^2$, leaving no other option (among finite-variance distributions).*

4 Why the Gaussian is Special

4.1 Fixed Point of Normalised Convolution

Define the operator $\mathcal{T}$: take independent copies X_1, X_2 of X and output $(X_1 + X_2)/\sqrt{2}$. In terms of characteristic functions: $\varphi_{\mathcal{T}X}(\omega) = \varphi_X(\omega/\sqrt{2})^2$.

Proposition 4.1. *$\mathcal{N}(0,\sigma^2)$ is the unique fixed point of $\mathcal{T}$ among finite-variance distributions.*

Proof. $\varphi(\omega) = e^{-\sigma^2\omega^2/2}$ satisfies $\big(e^{-\sigma^2(\omega/\sqrt{2})^2/2}\big)^2 = e^{-\sigma^2\omega^2/2}$. Uniqueness follows from Marcinkiewicz's theorem. □

In signal-processing language: a Gaussian pulse convolved with itself (and renormalised) stays Gaussian. Every other pulse shape drifts toward Gaussian under iteration.

4.2 Maximum-Entropy Distribution

Theorem 4.2 (Jaynes 1957 [5]). *Among all distributions on $\mathbb{R}$ with fixed mean μ and variance σ^2, the differential entropy $h(X) = -\int f \log f$ is uniquely maximised by $\mathcal{N}(\mu, \sigma^2)$, attaining $h = \frac{1}{2}\log(2\pi e\sigma^2)$.*

Sketch. Euler–Lagrange on $-\int f \log f$ with three Lagrange multipliers (normalisation, mean, second moment) gives $f^*(x) = \exp(\lambda_0 + \lambda_1 x + \lambda_2 x^2)$, which is Gaussian for $\lambda_2 < 0$. □

4.3 The Information-Theoretic Second Law

The Shannon–Stam inequality [6] states that entropy is monotone along the CLT sequence: $h(Z_{n+1}) \geq h(Z_n)$. Combined with the maximum-entropy theorem, the CLT has a thermodynamic reading: the normalised sum evolves irreversibly toward the maximum-entropy state, exactly as an isolated system evolves toward thermodynamic equilibrium.

5 Signal Processing: CLT as a Cascade of Filters

Identify the PDF f_i of each Y_i with an impulse response h_i. After n stages the combined response is $h_1 * \cdots * h_n$, the PDF of S_n. Rescaling time by $1/\sqrt{n}$ (equivalently, broadening the frequency axis by $\sqrt{n}$) gives Z_n, and the CLT says this converges to a Gaussian. **An arbitrary pulse shape, repeatedly self-convolved and renormalised, converges to a Gaussian pulse.**

In the frequency domain: repeated multiplication of the transfer function $H(\omega)$, followed by frequency stretching, squeezes the spectrum toward $e^{-\omega^2/2}$.

The Gaussian also saturates the Heisenberg–Gabor uncertainty principle $\Delta t \cdot \Delta\omega \geq \frac{1}{2}$, making it the *minimum-uncertainty waveform*. No other pulse is simultaneously as compact in both time and frequency.

6 Statistical Mechanics: Partition Functions as Convolutions

For a system of n independent subsystems with densities of states $g_i(\varepsilon)$, the total density of states is the n-fold convolution: $g_{\text{total}} = g_1 * g_2 * \cdots * g_n$. The canonical partition function factorises: $Z(\beta) = \prod_i Z_i(\beta)$. This is *convolution → product under the Laplace transform*—Table 2 in action.

The Maxwell–Boltzmann velocity distribution is a direct CLT consequence: each velocity component v_x is the cumulative result of many independent molecular collisions; by the CLT it converges to $\mathcal{N}(0, k_B T/m)$.

Boltzmann's entropy $S = k_B \ln \Omega$ is, in the continuum limit, Shannon's differential entropy up to a constant. Jaynes's maximum-entropy derivation of the canonical ensemble is therefore the same calculation as the max-entropy derivation of the Gaussian: maximise entropy subject to a fixed mean (energy or variance), obtain an exponential (Boltzmann or Gaussian) distribution.

7 Cross-Field Technology Transfer

The dictionary in Section 2 raises an ambitious question: *can a theorem proved with effort in one field be carried across the dictionary to yield a non-trivial new result in another?* This section argues the answer is yes, through six specific instances. For each we identify the source theorem, perform the translation, and discuss the degree to which the translated result is new.

7.1 Transfer 1: Stein's Lemma = Price's Theorem

Source (Probability).

Theorem 7.1 (Stein's Lemma [8]). *If $X \sim \mathcal{N}(0, \sigma^2)$ and g is absolutely continuous with $\mathbb{E}[|g'(X)|] < \infty$, then*

$$\mathbb{E}[X\, g(X)] = \sigma^2\, \mathbb{E}[g'(X)].$$

Stein's lemma is the engine behind Stein's method for distributional approximation, the SURE estimator, and several proofs of the CLT.

Translation. Differentiate the equation with respect to σ^2. On the left, use the fact that the distribution of X changes as σ^2 changes: $\frac{d}{d\sigma^2}\mathbb{E}[g(X)] = \frac{1}{2}\,\mathbb{E}[g''(X)]$.

Target (Signal Processing) — Price's Theorem [9].

Theorem 7.2 (Price 1958)**.** *For $X \sim \mathcal{N}(0, \sigma^2)$ and measurable g:*

$$\frac{d}{d\sigma^2}\,\mathbb{E}[g(X)] = \frac{1}{2}\,\mathbb{E}[g''(X)].$$

Price derived this independently in 1958 for the analysis of nonlinear devices driven by Gaussian noise. The two theorems are the same equation, derived a decade apart in different communities.

What transfers. Every result built on Stein's lemma (e.g., James–Stein shrinkage, Stein's method bounds on the CLT convergence rate) has a direct signal-processing proof via Price's theorem, and vice versa. In particular, the optimality of the SURE (Stein's Unbiased Risk Estimate) translates into a signal-processing statement about the minimum achievable MSE for any nonlinear estimator of a Gaussian signal.

7.2 Transfer 2: De Bruijn Identity = Gaussian Filtering = Heat Equation

Source (Information Theory).

Theorem 7.3 (De Bruijn Identity [7])**.** *Let $X_t = X + \sqrt{t}\,Z$ where $Z \sim \mathcal{N}(0,1)$ is independent of X. Then*

$$\frac{d}{dt}\,h(X_t) = \frac{1}{2}\,J(X_t),$$

where h is differential entropy and $J(f) = \int (f'/f)^2 f\,dx$ is Fisher information.

Translation. Adding Gaussian noise of variance t to X is identical to *convolving* f_X with the Gaussian kernel $\phi_t(x) = (2\pi t)^{-1/2} e^{-x^2/2t}$—the heat kernel at time t. The density f_{X_t} therefore satisfies the *heat equation*: $\partial_t f = \frac{1}{2}\partial_{xx} f$.

Target (Signal Processing + PDE). The combined statement is:

Gaussian filtering at width $\sqrt{t}$ $\equiv$ heat-equation evolution to time t $\equiv$ entropy increasing at rate $\frac{1}{2}J$.

What transfers. This triple equivalence has two immediately useful consequences not standard in signal processing textbooks:

1. *Fisher information is measurable by blurring.* Blur a signal f_X with Gaussians of increasing width $\sqrt{t}$ and track the rate of entropy change. The derivative at $t = 0$ equals $\frac{1}{2}J(X)$. This gives a purely signal-processing (filtering) algorithm for estimating the Fisher information of an amplitude distribution.

2. *Heat-equation scale-space analysis (Lindeberg 1994 [14]) is an entropy-theoretic framework.* The Gaussian scale-space, standard in computer vision, is exactly the curve of maximum-entropy-increase images, with Fisher information tracking the rate.

7.3 Transfer 3: Fisher Information Inequality ⇒ Bandwidth Irreversibility

Source (Information Theory).

Theorem 7.4 (Stam's Fisher Information Inequality [7])**.** *For independent X, Y:*

$$\frac{1}{J(X+Y)} \;\geq\; \frac{1}{J(X)} + \frac{1}{J(Y)}.$$

Translation. For a Gaussian distribution $\mathcal{N}(0, \sigma^2)$, the Fisher information is $J = 1/\sigma^2$, so the inequality reads $\sigma^2_{X+Y} \leq \sigma^2_X + \sigma^2_Y$, which is just Bienaymé's identity (equality for independent Gaussians). But for non-Gaussian noise sources, Stam's inequality is strictly tighter.

Via the dictionary: $J(f) \leftrightarrow$ (effective bandwidth)^2, because for a Gaussian the Fisher information equals the inverse variance, and variance in the time-domain corresponds to inverse bandwidth in the frequency domain.

Target (Signal Processing) — Bandwidth Budget.

Corollary 7.5 (Bandwidth Irreversibility)**.** *In a cascade of n independent additive noise stages with Fisher informations $J_1, \ldots, J_n$, the Fisher information of the total noise satisfies*

$$\frac{1}{J_{\text{total}}} \;\geq\; \sum_{i=1}^{n} \frac{1}{J_i}.$$

No downstream linear or nonlinear processing can increase J_{total}.

What is new. Standard noise-figure analysis (Friis formula) tracks average noise power: $N_{\text{total}} = N_1 + N_2 + \cdots$. This is additive in variance. Stam's inequality is a *strictly stronger* statement: it bounds the harmonic sum of Fisher informations, which for non-Gaussian noise sources gives a tighter lower bound on the effective noise floor than Friis alone. Moreover, because the data-processing inequality implies that Fisher information can only decrease under further processing [10], the degradation at each stage is *irreversible*. A signal processing engineer designing a multi-stage receiver has a Fisher-information budget that is strictly more constraining than a noise-power budget.

7.4 Transfer 4: Entropy Power Inequality ⇒ Superadditive Noise Floor

Source (Information Theory).

Theorem 7.6 (Shannon's Entropy Power Inequality [10])**.** *For independent X, Y with finite differential entropy:*

$$e^{2h(X+Y)} \;\geq\; e^{2h(X)} + e^{2h(Y)}.$$

Equivalently, with the entropy power $N(X) = \frac{1}{2\pi e} e^{2h(X)}$:

$$N(X+Y) \;\geq\; N(X) + N(Y).$$

Equality holds if and only if X and Y are both Gaussian.

Target (Signal Processing) — Superadditive Noise Floor. Identify X and Y with the amplitude distributions of two independent noise sources (e.g., thermal noise and quantisation noise). Then:

Corollary 7.7. *When two independent noise sources are combined additively, the* entropy power *of the combined noise is at least the sum of the individual entropy powers. The noise floor, measured in entropy power, is superadditive.*

What is new. The Friis noise-figure formula says the noise *power* is additive. The EPI says the noise *entropy power* is superadditive: $N(X+Y) \geq N(X)+N(Y)$, with strict inequality whenever either source is non-Gaussian. This means that mixing non-Gaussian noise sources (clipping noise, impulsive interference, quantisation noise) always produces a combined noise with higher entropy power than the naive additive prediction. The corollary provides a new fundamental lower bound on the achievable noise figure of any receiver that mixes non-Gaussian impairments. To our knowledge, this application of the EPI to multi-impairment noise floors is not standard in the signal processing or RF engineering literature.

7.5 Transfer 5: Berry–Esséen Theorem ⇒ Filter Symmetry Determines Convergence Speed

Source (Probability).

Theorem 7.8 (Berry–Esséen [3])**.** *Under the CLT hypotheses, with $\rho = \mathbb{E}[|Y|^3]$:*

$$\sup_x \left| F_n(x) - \Phi(x) \right| \;\leq\; \frac{C\,\rho}{\sigma^3 \sqrt{n}}.$$

The bound is tight: ρ/σ^3 is the absolute skewness, and it governs the $O(1/\sqrt{n})$ convergence rate.

Translation. Identify ρ/σ^3 with the *normalised third moment* (skewness) of the impulse response h of a filter. After n self-convolutions plus renormalisation, the cumulative response deviates from a Gaussian CDF by at most $C\,\mathrm{skew}(h)/\sqrt{n}$.

Target (Filter Design).

Corollary 7.9 (Symmetric Filters Converge Faster)**.** *A filter with a symmetric impulse response ($h(t) = h(-t)$, i.e., zero skewness and all odd moments zero) has its n-fold convolution converge to Gaussian at rate $O(1/n)$ rather than $O(1/\sqrt{n})$.*

The $O(1/n)$ rate for zero-skewness follows because the Berry–Esséen bound uses the third cumulant; when the third cumulant vanishes, the leading error term is governed by the fourth cumulant and the Edgeworth expansion [12] gives $O(1/n)$.

What transfers. This gives a concrete, quantitative design principle: if the application requires a cascade of n identical filters to behave as a Gaussian filter (e.g., for matched filtering, smoothing, or scale-space construction), use a *symmetric* impulse response. The convergence is quadratically faster in n. Conversely, if the cascade of an asymmetric filter is not yet Gaussian, the Berry–Esséen rate tells you exactly how many stages are needed.

7.6 Transfer 6: Cramér's Large-Deviations Theorem ⇒ Exponential Tail Decay of Cascaded Filters

Source (Probability).

Theorem 7.10 (Cramér's Theorem [13])**.** *Let $I(a) = \sup_{t\in\mathbb{R}}\big(ta - \Lambda(t)\big)$ where $\Lambda(t) = \log M_X(t) = \log \mathbb{E}[e^{tX}]$ is the log-MGF. Then for $a > \mathbb{E}[X]$:*

$$\lim_{n\to\infty} \frac{1}{n} \log P\left(\frac{S_n}{n} \geq a\right) = -I(a).$$

I is the Legendre–Fenchel transform of Λ, also called the rate function.

Translation. Under Table 2: $M_X(t) = \mathbb{E}[e^{tX}]$ is the Laplace transform of f_X evaluated on the *real axis* $t \in \mathbb{R}$. In signal processing this is $H(s)\big|_{s=t\in\mathbb{R}}$, the transfer function restricted to the real axis (as opposed to the imaginary axis used in Fourier analysis).

Target (Signal Processing) — Tail Decay Rate of a Cascade.

Corollary 7.11 (Rate Function from Transfer Function)**.** *Let $H(s)$ be the bilateral Laplace transform of a filter impulse response h. After n self-convolutions followed by time-scaling (i.e., rescaling to unit mean), the probability that the filter output exceeds a decays as*

$$P(\mathit{output} \geq a) \asymp e^{-n\,I(a)}, \qquad I(a) = \sup_{t\in\mathbb{R}}\big(ta - \log H(t)\big).$$

The exponential tail-decay rate is the Legendre transform of the log-transfer-function evaluated on the real axis.

What transfers. For a Gaussian filter, $\log H(t) = \sigma^2 t^2/2$, giving $I(a) = a^2/(2\sigma^2)$: Gaussian tails. For any other filter, $I(a)$ is strictly smaller than the Gaussian rate function in the non-Gaussian directions, quantifying exactly how far the n-fold cascade still is from Gaussian convergence.

This has two practical consequences:

1. **Bit-error rate analysis.** In digital communications, the probability of a symbol error is the tail probability of the noise distribution after matched filtering. Cramér's theorem gives the *exact exponential decay rate* of this error probability in terms of the real-axis transfer function of the filter, without requiring the CLT approximation (Gaussian tails) to hold.

2. **Detecting non-Gaussianity.** If the measured tail of a cascaded-filter output decays faster than $e^{-nI_{\mathrm{Gauss}}(a)}$, the individual filter response is super-Gaussian (lighter-tailed than Gaussian). If it decays slower, the response is sub-Gaussian (heavier-tailed). The rate function is thus a quantitative fingerprint of non-Gaussianity, inherited directly from probability theory with no separate derivation required.

8 Transforms: A Unified View

All transforms in this paper share: $(f * g)^\wedge(\xi) = \hat{f}(\xi) \cdot \hat{g}(\xi)$. The reason is that exponential functions $e^{i\omega x}$ are eigenfunctions of the convolution operator with eigenvalue $\hat{f}(\omega)$.

Taylor series ↔ Moments. $M_X(s) = \sum_{k \geq 0} \mathbb{E}[X^k]\, s^k/k!$: the moments are precisely the Taylor coefficients of the MGF. A distribution is (under Carleman's condition) uniquely determined by all its moments, just as an analytic function is determined by its Taylor series. The Fourier transform translates this local characterisation (moments = local derivatives) into a global one (spectral amplitudes), and vice versa.

Complex-analytic rigidity. The characteristic function $\varphi_X(\omega)$ extends to an analytic function. Marcinkiewicz's theorem uses this: an analytic function satisfying $\varphi(\omega)^n = \varphi(\sqrt{n}\,\omega)$ for all n must be $e^{-c\omega^2}$. This is the complex-analytic proof that the Gaussian is the only stable distribution with finite variance.

9 Discussion and Open Questions

Summary of cross-field transfers. The six transfers in Section 7 share a common structure: a theorem that required deep effort in its home field (Stein 1956, De Bruijn 1959, Stam 1959, Shannon 1948, Berry–Esséen 1941–1956, Cramér 1938) translates immediately into a new result in signal processing. The translations are not trivial because the fields have developed incompatible notation and intuition; the dictionary in Section 2 provides the Rosetta stone. To the best of our knowledge, Transfers 3, 4, and 6 have not appeared in signal processing textbooks.

Is any of this "obvious"? The individual equations are in many cases the same (e.g., Stein = Price). What is non-obvious is that: (a) the correspondence is exact rather than approximate, and (b) the transferred result is non-trivial in the target field. A signal processing engineer who knows Friis's formula does not automatically know the EPI-based superadditive noise floor; a statistician who knows Stein's lemma does not automatically recognise it as Price's theorem. The novelty lies in *making the transfer explicit*.

Open Question 1: Non-Gaussian attractors and q-deformed CLT. The CLT has Gaussian as its unique attractor for finite-variance distributions. For distributions with infinite variance (α-stable distributions, Lévy flights), the attractor is the family of α-stable laws whose characteristic function is $e^{-|\omega|^\alpha}$. These are the fixed points of the Fourier operator in the heavy-tail regime. In non-extensive statistical mechanics [15], the role of the Gaussian is played by the q-Gaussian, maximising Tsallis entropy instead of Shannon entropy. The corresponding CLT (Umarov–Tsallis–Steinberg 2008) gives q-Gaussians as attractors. *What is the signal-processing analogue of the q-Gaussian filter cascade? Is there a q-deformed convolution theorem?*

Open Question 2: Free probability and non-commutative signal processing. In random matrix theory, the CLT for independent large matrices gives Wigner's semicircle law (not Gaussian) via the *free convolution* (Voiculescu). In large MIMO wireless systems, the singular value distribution of the channel matrix follows the Marchenko–Pastur law, a manifestation of the free CLT. The free CLT fails precisely because matrix operations do not commute. *Is there a non-commutative signal processing, applicable to matrix-valued signals or MIMO channels, governed by free convolution? What would a "free matched filter" look like?*

Open Question 3: Thermodynamic Uncertainty Relations as SNR bounds. Recent non-equilibrium statistical mechanics has established the Thermodynamic Uncertainty Relation (TUR) [16]: for any steady-state current J in a system producing entropy at mean rate $\dot{\Sigma}$, $\mathrm{Var}(J)/\langle J \rangle^2 \geq 2k_B/\dot{\Sigma}$. Via the dictionary, $J \leftrightarrow$ signal output, $\dot{\Sigma} \leftrightarrow$ information loss rate. *Does the TUR translate into a*

fundamental lower bound on the coefficient of variation of the output of any nonlinear information-processing system, expressed in terms of how much entropy the system produces? If so, more irreversible (higher-entropy-production) systems could achieve better SNR—a counterintuitive design principle.

Open Question 4: t-distribution and finite thermodynamic systems. The CLT is asymptotic. For finite n, the normalised sum has heavier tails, described by Student's t_ν with $\nu = n - 1$. In statistical mechanics, the thermodynamic limit $N \to \infty$ corresponds to $n \to \infty$ in the CLT. A "gas of 5 molecules" should exhibit t-distributed energy fluctuations. *Is this consistent with results from quantum statistical mechanics of few-body systems and the eigenstate thermalisation hypothesis [17]?*

10 Conclusion

Convolution is not merely a shared notation but a shared *structure*. The Central Limit Theorem unifies as: a fixed-point theorem (analysis), a filter-cascade limit (signal processing), a maximum-entropy attractor (information theory), and a thermodynamic equilibrium (statistical mechanics).

The deeper payoff of the correspondence dictionary is that it lets theorems travel. We have shown six examples where a deep result in probability or information theory translates, with no new mathematics, into a non-trivial and in some cases previously unformulated result in signal processing. The Entropy Power Inequality gives a superadditive noise floor; the Fisher Information Inequality gives bandwidth irreversibility beyond Friis; Berry–Esséen gives a design rule for fast-converging filter cascades; De Bruijn makes Gaussian filtering equivalent to the heat equation and Fisher information measurable by blurring; Cramér gives the exact exponential tail rate of any cascaded filter from its real-axis transfer function; and Stein's lemma equals Price's theorem.

The four open questions point toward richer correspondences: q-deformed convolutions, non-commutative signal processing via free probability, TUR-based SNR bounds, and finite-n analogues in few-body quantum systems.

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